National AI Hackathon 2026 by SchubKI

16-21 of October 2026 (16 – opening, 17-18 – work, 19-20 – evaluation, 21 - closing)

Constructor Technology

Challenge name

Mind Reading Using Machine Learning and EEG

Challenge description

Electroencephalogram (EEG) data is a relatively inexpensive biological signal that can be easily collected using simple and non-invasive procedures. Despite a relatively low signal-to-noise ratio, EEG has proven useful for a wide range of research and real-world applications.

This Hackathon will explore the use of state-of-the-art machine learning approaches (such as dimensionality reduction, video foundation models, and world models) to perform a simple form of **"mind reading"**. Participants will work with EEG recordings collected while subjects viewed short three-second videos. The goal is to be able to tell **which video a subject was viewing based solely on their EEG activity**.

More specifically, given an EEG recording and a set of candidate videos, participants will develop a model that ranks the candidate videos from most to least likely to have elicited the recorded brain activity.

Beyond predictive performance, the challenge aims to encourage the integration of machine learning expertise and cognitive-psychology domain knowledge. We are particularly interested in approaches that not only are accurate but also provide scientifically meaningful insights into the similarities and differences between how biological and artificial systems process visual stimuli.

To motivate scientifically grounded and domain-appropriate reasoning instead of over-engineering and dataset hacking, the challenge comprises two groups of subjects: four development subjects and two partially held-out subjects. For all six subjects, training recordings are provided together with the video that elicited each recording. For the four development subjects, the labels of the test recordings are also provided for validation and model selection. For the two partially held-out subjects, the test labels are withheld and used to evaluate the generalizability and robustness of the solutions. All participating teams will be required to share the training pipelines and model weights to enable the reproduction and assessment of the proposed approaches.

The solutions are required to be written in Python 3.11. Template notebooks for loading and processing the EEG data, as well as for training a baseline model, are provided by the organizers.

Challenge (input + details)

The dataset contains EEG recordings from six subjects (sub-01 to sub-06), collected while they viewed short videos of approximately three seconds in duration. The videos used as stimuli are also provided. For all six subjects, the correspondence between each training recording and its stimulus video is provided. For the four development subjects (sub-01 to sub-04), this correspondence is also provided for the test recordings, which can therefore be used to validate models. For the two held-out subjects (sub-05 and sub-06), the test recordings and the set of possible stimulus videos are provided, but the correspondence between test recordings and videos is withheld.

Participants are expected to use their trained models to rank, for each test recording of sub-05 and sub-06, the 102 candidate videos from most to least likely to have elicited the recorded brain activity (see Expected output).

Participants are free to design their own preprocessing, feature extraction, and prediction pipelines. Possible approaches include traditional feature engineering, dimensionality reduction, time-series modeling, deep neural networks, video foundation models, and other multimodal or representation-learning approaches. Since the videos were presented with their sound, audio and audio-visual models are also relevant.

The challenge input comprises:

- stimuli_train: a folder containing the 1,000 videos (0001.mp4 to 1000.mp4) used as stimuli for the training EEG recordings. Not all subjects have EEG recordings for every video (850 to 972 recordings per subject).
- stimuli_test: a folder containing the 102 videos (1001.mp4 to 1102.mp4) used as stimuli for the test EEG recordings. All subjects viewed the same 102 test videos.
- trn: a folder containing the training EEG recordings, one file per subject (sub-XX_trn.npz). Each file is a dictionary in which each key is the ID of a stimulus video, written without leading zeros (e.g., key "35" corresponds to stimuli_train/0035.mp4), and the associated value is a 2D array of shape 17 x 1500: 17 EEG channels sampled at 500 Hz for three seconds, starting at video onset.
- tst: a folder containing the test EEG recordings, one file per subject (sub-XX_tst.npy). Each file contains an array of shape 102 x 17 x 1500, i.e., one recording per test video, in shuffled order.

- tst_stim: a folder containing the ground-truth stimulus IDs of the test recordings (sub-XX_tst_stim.npy): an array of 102 video IDs between 1001 and 1102, in the same order as the recordings in the corresponding tst file. These files are provided only for sub-01 to sub-04; the files for sub-05 and sub-06 are withheld and used by the organizers for evaluation.

For convenience, the organizers provide two Python notebooks as starting points:

- 01_preprocess_and_train_model.ipynb: uses the development data to build a pipeline that, given an EEG recording and a set of candidate videos, identifies the stimulus most likely to have elicited the recorded brain activity.
- 02_predict_movies.ipynb: Loads the model trained in the previous notebook and applies it to the held-out EEG recordings to predict the corresponding stimulus videos.

Minimal requirements

- A valid solution must be implemented in **Python 3.11**.
- Participants may use additional datasets and/or pre-trained models, provided that they are **publicly available and accessible to all participants** and were released before the start of the hackathon. For example, participants may use publicly released open-source time-series models or publicly available pre-trained foundation models.
- Do not use any data from the test set of any subject to train or fine-tune your model. The training data of all six subjects may be used for training. The test recordings and labels of sub-01 to sub-04 may be used only for validation and model selection; in particular, they must not be used as reference responses (templates) for identifying the test recordings of sub-05 and sub-06. Unsupervised normalization of the test recordings and of the video features (e.g., z-scoring) is allowed.
- Participants must generate predictions for the held-out test recordings of sub-05 and sub-06 in the format described under Expected output.
- Participants must submit a report.
- A reproducible training and inference pipeline must be uploaded to the designated GitHub repository (without the videos or extracted frames).
- Participants must provide the source code and model weights necessary to reproduce their submitted solution. Weights of public pre-trained models do not need to be uploaded, but their exact name and version must be given.
- Presentation (max. 5 minutes)

Expected output

The expected output is a model and inference pipeline that, given an EEG recording and a set of candidate videos, ranks the videos from most to least likely to have elicited the recorded brain activity. For the held-out test set, participants will use this pipeline to rank the 102 candidate videos for each test recording of sub-05 and sub-06.

The predictions must be submitted as two files, **sub-05_ranked_candidate_list.npy** and **sub-06_ranked_candidate_list.npy**, each containing an integer array of shape 102 x 102 saved with ***numpy.save***. Row *i* corresponds to the *i*-th recording in ***tst/sub-XX_tst.npy*** (in the original, unsorted order) and must contain all 102 video IDs (1001 to 1102, not the column indices 0 to 101), ordered from most to least likely. This is the format (shape and content) of the ranked array created in **02_predict_movies.ipynb**; in the notebook, however, the rows follow the label-sorted order used for validation, **so do not reorder the recordings when building the submission**. **Files with a different name, shape, or content cannot be scored.** To check the format, run the same submission code on **sub-01** and score the result with **tst_stim/sub-01_tst_stim.npy.**

The final submission must include:

- **Source code (mandatory) - a Git repository link:**
  - the Python source code,
  - the training and inference pipeline,
  - all model weights required to reproduce the results (for public pre-trained models, their exact name and version),
  - predictions for the held-out test set
- **Presentation (max. 5 minutes):** approach, diagrams, results, failure cases and lessons learned
- **Slides:** approach, diagrams, results, failure cases and lessons learned

The complete submission should allow the organizers to reproduce and evaluate the reported results.

Evaluation criteria

Quantitative (70%)

For each subject, the dataset is divided into training and test splits. Each split contains EEG recordings along with the corresponding videos presented as stimuli during EEG acquisition.

The task is straightforward: **given the EEG recordings in the training split, develop a pipeline that predicts which of the 102 videos evoked each of the 102 EEG recordings in the test split**. The complete training and test EEG data are available for all six subjects. However, the ground-truth correspondence between test recordings and stimuli (i.e., the correct stimulus order) is only provided for four of the six subjects.

For the last two subjects (sub-05 and sub-06), participants are expected to infer the stimulus order using the pipeline trained on the training data and to submit their ranking of the 102 candidate videos for each of the 102 test recordings. The performance of all hackathon participants will be determined by the top-1 accuracy (the fraction of the 102 test recordings whose highest-ranked video is the correct video; chance level $1/102 \approx 1.0\%$) and the top-5 accuracy (the fraction of test recordings whose correct video is among the five highest-ranked candidates; chance level $5/102 \approx 4.9\%$).

Qualitative (30%):

- Technical soundness and novelty of the approach (15%)
- Engineering quality and reproducibility of the code (10%)
- Presentation, including analysis of failure cases and limits (5%)

Dataset links

- **Materials download:** the link
- **Video intro for the Challenge here:** the link